Ans: B

2 The Rayleigh-Jeans Law is an approximation to the spectral radiance of electromagnetic radiation as a function of wavelength from a black body at a given temperature through classical arguments. For wavelength λ, it is: $B=\dfrac{2ckT}{\lambda^4}$, where c is the speed of light, k is the Boltzmann constant and T is the thermodynamic temperature.

What is the unit of B, expressed in SI base units?

A $\mathrm{m^{-3}\ s^{-1}\ K}$

B $\mathrm{m^{2}\ s^{-3}\ kg}$

C $\mathrm{m^{-3}\ s^{-3}\ kg}$

D $\mathrm{m^{-1}\ s^{-3}\ kg}$

$$[B]=\frac{\mathrm{m\ s^{-1}\ J\ K^{-1}\ K}}{\mathrm{m^4}}$$

$$[B]=\mathrm{m^{-3}\ s^{-1}\ J}$$

$$[B]=\mathrm{m^{-3}\ s^{-1}\ kg\ m^{2}\ s^{-2}}$$

$$[B]=\mathrm{m^{-1}\ s^{-3}\ kg}$$

Ans: D

3 Two identical blocks are released from rest from the top of two ramps as shown below.